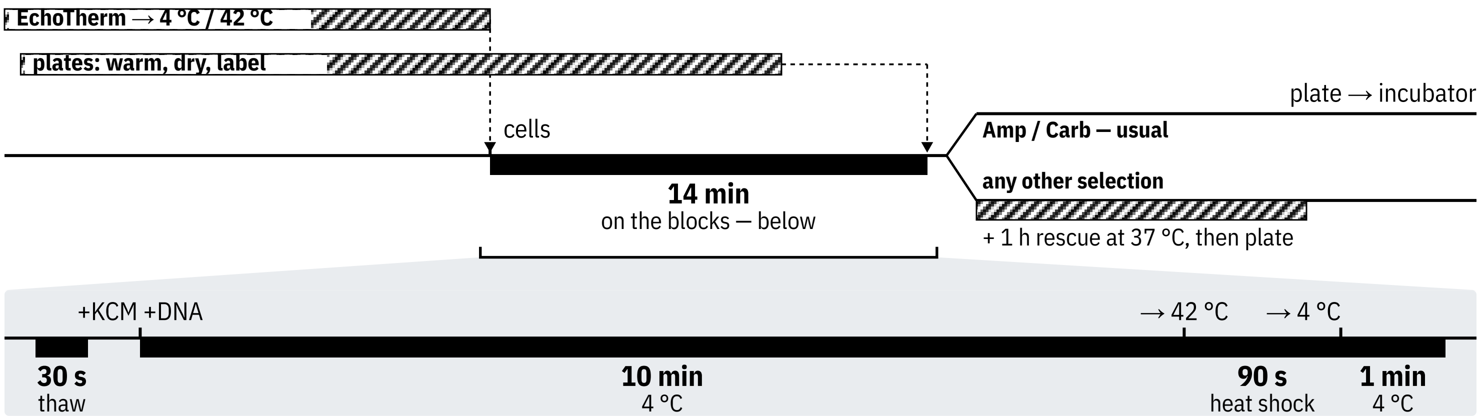
KCM Heat-Shock Transformation

TIMING



PROTOCOL

1. **Start the slow things first.** Get your plates — check the antibiotic(s) on your labsheet — and warm them in the **incubator**. Cold plates carry condensation and **you cannot write on them**.
2. Turn on the **EchoTherm**: **block A 4 °C**, **block B 42 °C**. Leave both at temperature throughout.
3. Once warm and dry, **label the bottom** of each plate: date, initials, strain, plasmid, selection.
4. Put a **100 µL** cell aliquot and the DNA tube on block A. Thaw ~30 s, add **25 µL KCM** to the cells, pipette gently. One aliquot does **3 reactions**.
5. Add **40 µL** of the cell/KCM mix to the DNA. Mix gently.
   – Assumes DNA is ~20% of the total; **too much dilutes the salts**. Large reaction: the whole tube of cells. Retransforming a miniprep: **0.5 µL** into 10 µL cells.
6. **Run the block sequence on the timeline above.**
7. **Rescue only if the selection is not Amp/Carb:** add **200 µL 2YT**, move to a 1.5 mL tube, shake 37 °C for **1 h**.
8. Plate everything. Incubate **inverted** at 37 °C overnight, and **cancel the temperature programs**.